

Lab 1 – TrashTag Product Description

Adam T. Daif

Old Dominion University

CS411W

Professor Hosni

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1 Introduction

Litter in outdoor recreational areas, such as rivers, parks, lakes, and greenways, remains an environmental and operational concern for communities. Trash discarded by visitors or illegally dumped not only disrupts natural ecosystems but also poses safety hazards, economic burdens, and long-term environmental impairment. According to the U.S. Environmental Protection Agency, aquatic trash harms ecosystems through entanglement, ingestion, and habitat disruption, contributing to the growing global problem of microplastic pollution (EPA, 2023).

Municipalities also bear the significant financial burden; U.S. cities collectively spend over \$11.5 billion annually on litter cleanup alone (Keep America Beautiful, 2020). Beyond the environmental and financial impact, the current process of identifying and removing trash is inefficient. Cleanups often depend on volunteers or city workers walking long distances with little guidance, hoping to encounter problem areas. Citizens who find large items (such as tires, furniture, or appliances) often do not know whom to notify or how urgently cleanup teams will be able to respond. This contributes directly to several critical gaps.

This inefficiency contributes to several critical gaps in the current cleanup process. Cleanup organizations struggle to quickly and accurately identify illegal dumping sites, often deploying labor without reliable, current information about the location or scale of debris. At the same time, citizens lack a structured and accessible channel for reporting litter, resulting in vague or delayed reports that limit organizational response effectiveness. Moreover, individuals typically encounter trash while engaged in other activities such as hiking or swimming, and once they leave the area, the opportunity or motivation to report the issue diminishes. Finally, when there is no visible acknowledgment of participation or measurable evidence of impact,

individuals may feel that their contributions are insignificant, reducing long-term engagement in cleanup efforts.

An effective solution must therefore provide an accessible, real-time reporting mechanism that captures precise location data, establish a direct communication channel between citizens and cleanup organizations, minimize the effort required to participate, and offer visible feedback that reinforces the value of individual contributions. TrashTag is designed to address each of these needs through a geolocation-based reporting platform that allows users to upload geotagged photographs of trash directly from their devices. These reports are displayed in real time on an interactive map, enabling organizations to identify hotspots, assess the scope of cleanup efforts, allocate appropriate resources, and coordinate volunteers more efficiently.

To sustain user engagement over time, TrashTag incorporates gamification elements such as points, reporting streaks, and visible cleanup milestones that reward participation and directly address the perceived impact gap. TrashTag is a unified data management and communication platform that enables individuals to contribute meaningfully to environmental cleanup efforts, amplifies the efforts of environmental organizations, and streamlines the overall coordination process. By transforming casual public observation into actionable environmental data, TrashTag provides an organized and scalable solution to a problem that is otherwise dependent on inconsistent reporting and manual discovery.

2 Product Description

TrashTag is a web-based reporting and cleanup coordination tool intended to help communities identify and clean up trash and illegal dumping in their natural and recreational areas. It is a tool that allows users to upload images of the trash they see, automatically tag their own location, and share those reports on a map. Cleanup organizations have access to these

reports and are able to claim and clean them up, making the process more efficient with less time and resources. The purpose and goal of the TrashTag tool is to reduce pollution in the environment and increase safety and transparency with a more organized approach to cleaning up the environment.

2.1 Key Product Features and Capabilities

TrashTag also provides users with the opportunity to report trash by uploading a photo. The photo will be stamped with the location and date of the photo. The reported trash will be displayed on a map, and users can easily see the reported trash and the extent of the problem. Cleanup organizations can claim reported locations within the system and update the status once debris has been removed, providing transparency and accountability throughout the process.

The uniqueness of TrashTag can be identified in the way it provides users with the opportunity to make casual observations and transform them into useful data. Unlike other methods of reporting trash, which rely on word of mouth, phone calls, and emails, TrashTag provides users with the opportunity to make real-time and location-based reports. This way, users can be sure of the location of the trash and the efforts needed to clean it up. This will prevent duplication of efforts and wasted resources.

2.2 Major Components (Hardware/Software)

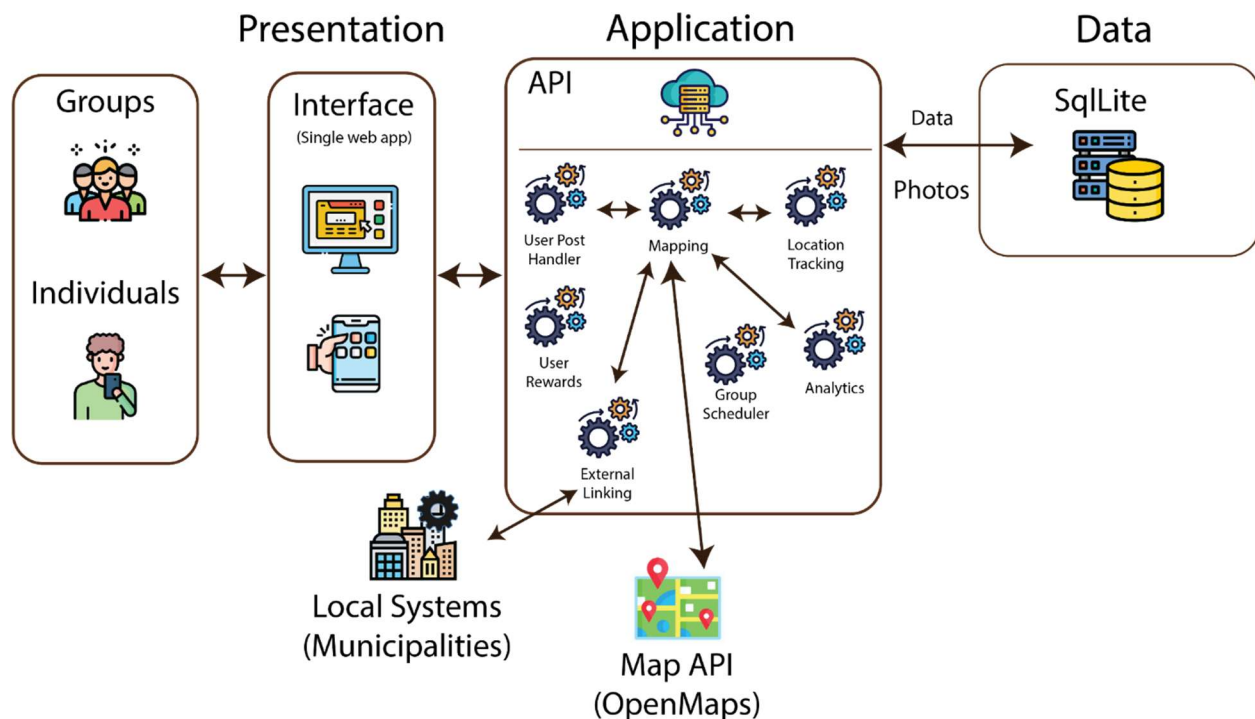
TrashTag is designed as a web application, so no specialized hardware is required beyond a standard internet-enabled device. Users and cleanup organizations access the system through a normal browser, while the backend runs on cloud-hosted servers that store data, process images, and manage authentication.

As illustrated in the Major Functional Component Diagram (MFCD) shown in Figure 1, TrashTag is structured into three primary layers. The model consists of the database that stores

users, trash reports, photos, events, organizations, and cleanup records. The functional layer includes the business logic that handles photo uploads, EXIF data extraction, map rendering, report claiming, and point tracking. The component layer consists of the web interface, the REST API, the database engine, and cloud-based image storage.

Figure 1

TrashTag Major Functional Component Diagram



The software stack includes a JavaScript-based frontend for the user interface, a Django backend with ORM-based data management, a SQLite database (since it is embedded into Django by default) with geographic data support, and cloud storage for images. Together, these components provide a scalable, secure, and responsive platform capable of handling real-time reporting and mapping.

3 Identification of Case Study

TrashTag is being designed for people who frequently visit outdoor recreational sites like rivers, parks, and trails, as well as organizations that are tasked with keeping these sites clean. This includes environmental non-profits, clean-up groups, and local government bodies. The need for TrashTag arises from the fact that these groups do not have an effective means of identifying and coordinating trash clean-up efforts.

The initial case study group will include a small number of local volunteers and clean-up groups who will test the TrashTag prototype by submitting geotagged photo reports of trash, viewing those reports on the interactive map, receiving real-time notifications, and gaining points for activity as part of TrashTag's gamification system. Feedback from this group will help to shape the features and functionality of the system.

For example, a volunteer visiting a local park notices a discarded tire near a riverbank. Using the TrashTag platform on their phone, the volunteer takes a photograph of the trash and uploads it through the reporting interface. The system automatically extracts the geolocation data from the image and places a report marker on the interactive map. A nearby cleanup organization member monitoring the map receives a real-time notification about the newly reported trash location. He reviews the report, claims the cleanup task for his team, and schedules a cleanup activity. Once the trash has been removed, the organization updates the report status to indicate that the issue has been resolved. The volunteer who originally reported the trash and the cleanup organization's participating members are all awarded participation points through the platform's gamification features.

TrashTag could potentially be used by city governments, park departments, schools, environmental organizations, and even property owners in the future. The system has the

potential to be used on a large scale for environmental monitoring and data-informed policy-making.

4 Glossary

EXIF (Exchangeable Image File Format): Metadata embedded within digital images that stores information such as the date, time, and GPS coordinates of the image capture.

Gamification: The incorporation of game-like mechanics such as rankings and achievement systems to motivate user participation and increase engagement in cleanup operations.

Geotagging: The addition of GPS coordinates to images that enable precise location identification where trash was reported.

Illegal Dumping: The unauthorized disposal of trash and other hazardous materials onto private and/or public property.

Live Real-Time Updates: Live data synchronization displaying trash reports to both users and cleanup organizations instantaneously.

Major Functional Component Diagram (MFCD): A diagram that illustrates the relationships between the major hardware, software, and system components within an application.

Metadata: Descriptive information within data files such as date, location, and timestamp data in photos to be submitted to TrashTag.

Representational State Transfer (REST): An architectural style for designing networked applications using standard HTTP methods to allow communication between client and server systems.

5 References

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